

Lead Enrichment Pipeline

Event-driven enrichment service handling identity resolution, fuzzy dedupe, and bidirectional CRM sync with conflict reconciliation.

Go	Kafka	Postgres	Salesforce API
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VERSION	STATUS	MODE	GO	OWNER	UPDATED
4.0.1	Complete	Maintenance	1.23	RevOps Eng	Sep 2026

1. Overview

The Lead Enrichment Pipeline turns raw inbound leads into deduplicated, enriched, CRM-resident records without human triage. Every lead source — forms, event lists, vendor feeds, product signups — publishes to a Kafka topic in a common envelope. The pipeline normalises the payload, resolves it against existing people and accounts, fills gaps from enrichment vendors, and writes the result to Salesforce.

Identity resolution is the hard part and the reason the service exists. The same human arrives as `j.smith@acme.com`, `jsmith@acme.io`, and a LinkedIn export with a personal address; the matcher collapses these into one record with a deterministic, auditable decision trail, and routes ambiguous cases to a review queue instead of guessing.

2. Event flow

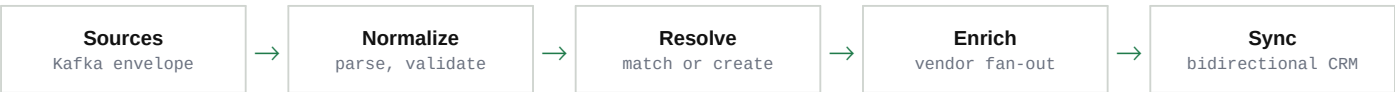


Figure 1 — Each stage commits its offset independently, so a vendor outage stalls enrichment without blocking normalisation or CRM sync.

Processing is idempotent by design. Each event carries a stable source key, and every stage writes under that key, so replaying a partition after an outage converges to the same state rather than creating duplicates. This property is what allows the pipeline to be restarted freely during incidents.

3. Pipeline stages

Ingest	Kafka consumer groups per source, with a dead-letter topic for malformed envelopes.
Normalizer	Email and phone canonicalisation, domain extraction, free-form company name cleanup.
Matcher	Blocking on domain and normalised name, then similarity scoring on the candidate set.
Enricher	Vendor fan-out with per-vendor budget, cache, and a fallback order by field coverage.
Sync worker	Bidirectional Salesforce sync with field-level change tracking on both sides.
Reconciler	Applies the conflict policy and records every rejected value for audit.

4. Match thresholds

SIGNAL	WEIGHT	BEHAVIOUR
Exact email	Decisive	Immediate merge; no further scoring.
Domain + name similarity	0.55	Jaro-Winkler above 0.92 on normalised names.
Phone match	0.25	E.164 normalised; mobile only, shared lines excluded.
Combined score	≥ 0.80 merge	0.60–0.79 routes to review; below 0.60 creates new.

5. Event envelope

```
{
  "source":      "webform.pricing",
  "source_key":  "wf-2026-09-11-8831",  // idempotency key
  "observed_at": "2026-09-11T17:04:22Z",
  "person": {
    "email": "J.Smith@Acme.com",
    "name":  "Jim Smith",
    "title": "VP Engineering"
  },
  "company": { "name": "Acme, Inc.", "domain": "acme.com" },
  "consent": { "basis": "legitimate_interest", "region": "US" },
  "enrich":  ["firmographics", "technographics"]
}
```

Listing 1 — Sources own only the envelope. Normalisation, casing, and formatting are the pipeline's job.

6. Conflict reconciliation

- **Precedence.** Field-level rules decide which side wins: human CRM edits beat vendor data, vendor data beats form input, and form input beats an older vendor value. Precedence is configured per field, not per record.
- **Detection.** Both sides carry a change watermark, so a value edited in Salesforce since the last sync is recognised as a genuine conflict rather than silently overwritten.
- **Audit.** Every rejected value is written to a conflict log with the rule that rejected it, which makes a suspicious field change answerable in one query.

7. Throughput

DIMENSION	MEASURED	NOTES
Sustained ingest	9,400 events/min	Twelve consumers, 4 vCPU each.
End-to-end latency	2.4 s p50 / 7.1 s p95	Ingest to CRM write, cache warm.
Auto-merge rate	93.1%	Remainder splits between review queue and new records.
False merge rate	0.12%	Sampled audit of 2,000 merges per quarter.

8. Operations and constraints

- Salesforce API budget is the binding limit, not compute. The sync worker batches to composite requests and backs off on the daily quota, spilling to a retry topic rather than dropping work.
- Consent basis travels with the record and gates enrichment: leads in regions requiring explicit consent skip vendor lookup entirely until consent is recorded.
- Vendor spend is capped per day and per field. When a budget is exhausted the enricher degrades to cached and derived values instead of failing the event.

9. Maintenance status

AREA	STATE	COMMITMENT
Feature work	Closed	Scope complete as of 4.0; no new capability planned.
Security and deps	Active	Monthly dependency updates and CVE patching.
Vendor adapters	Active	Breaking vendor API changes fixed within one sprint.